Recent Perspectives on the Relations between Fecal Mutagenicity, Genotoxicity, and Diet

DNA damage is an essential component of the genesis of colonic cancer. Gut microbial products and food components are thought to be principally responsible for the damage that initiates disease progression. Modified Ames tests and Comet assays have been developed for measuring mutagenicity and genotoxicity. Their relevance to oncogenesis remains to be confirmed, as does the relative importance of different mutagenic and genotoxic compounds present in fecal water and the bacteria involved in their metabolism. Dietary intervention studies provide clues to the likely risks of oncogenesis. High-protein diets lead to increases in N-nitroso compounds in fecal water and greater DNA damage as measured by the Comet assay, for example. Other dietary interventions, such as non-digestible carbohydrates and probiotics, may lead to lower fecal genotoxicity. In order to make recommendations to the general public, we must develop a better understanding of how genotoxic compounds are formed in the colon, how accurate the Ames and Comet assays are, and how diet affects genotoxicity.